Organic Chemistry Cumulative Exam
January 22, 2009

A recent article in Science, arguably the leading science journal in the world, entitled "Reactome Array: Forging a Link Between Metabolome and Genome," contained the equation shown on the following page. Chemists identified problems with the scheme, and it was later disclosed that no synthetic organic chemists reviewed the article. The editors of Science have published "an editorial expression of concern" about the article.

Identify the problem(s) with the chemical structure(s), the indicated reaction and any other feature of the scheme and/or caption.

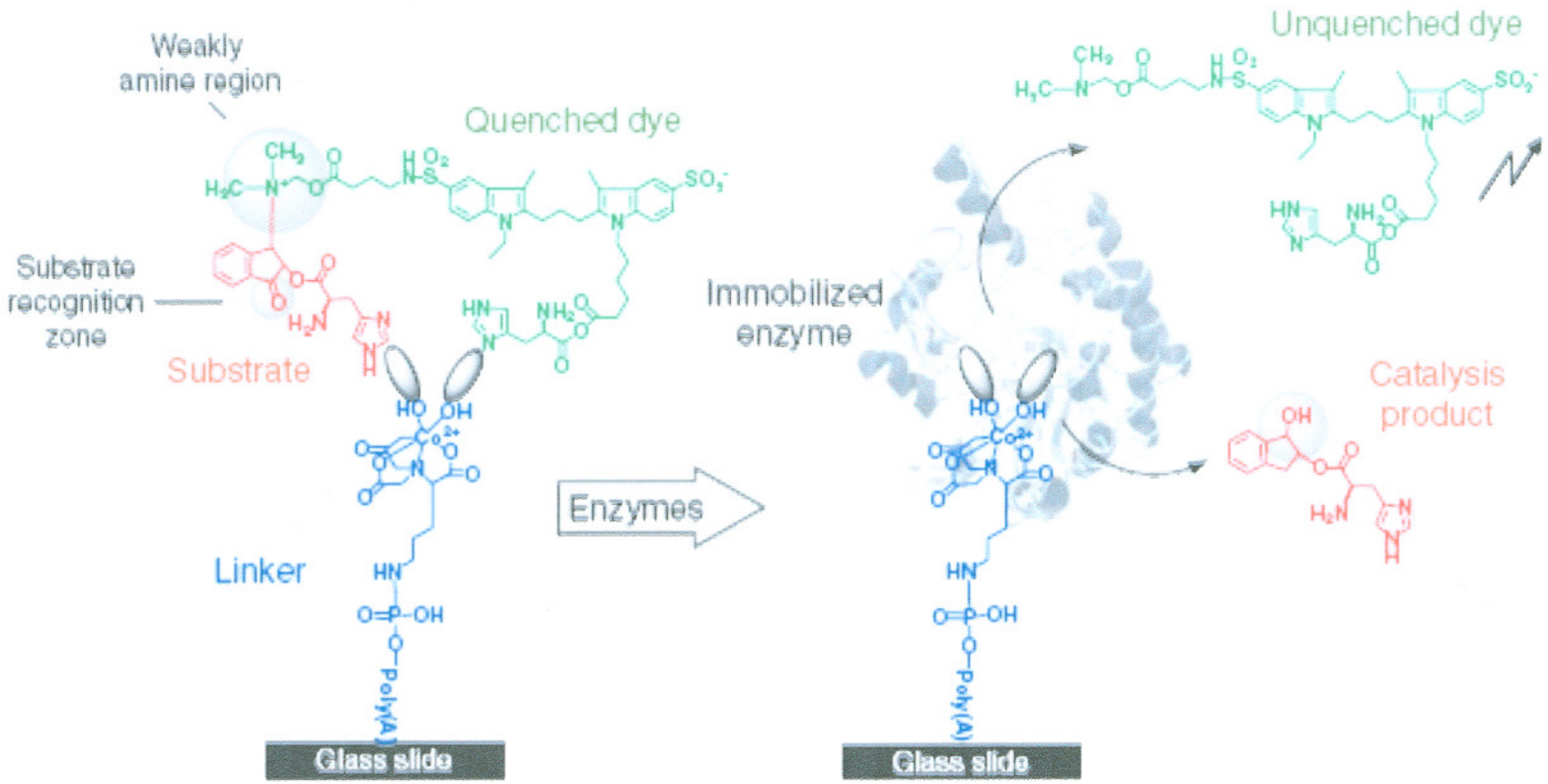

Fig. 1. The reactome strategy. The generic structure of reactome metabolites involves three linked components: the enzyme substrate-metabolite, the quenched dye, and the linker used to immobilize the complex on the array or on nanoparticles. The substrate-metabolite is linked to the quenched dye through a labile nitrogen bond, and both the dye and the substrate are anchored to the Co(II)-containing poly(A) linker by histidine "tags." Details of the synthetic strategy are provided in fig. S1. An enzyme-catalyzed chemical change in the substrate at a position adjacent to the weakly amine region causes rupture of the labile nitrogen:metabolite bond and release of the quenched Cy3 dye. This in turn provokes release of the reaction product and the histidine "tags" anchored to the Co(II), thereby exposing an active cobalt cation that ligates and immobilizes the enzyme on the array spot. The released dye is no longer quenched and gives a fluorescent signal. The nature of the reaction and the catalysis product is defined by the position to which the quenched dye and the substrate are linked (table S2).